Discrepancy of the Complete Set System

Theorem 1. *Let X be a finite set with $|X| = n$, and let $\mathcal{F} = \mathcal{P}(X)$ be the complete set system on X, i.e. the collection of all subsets of X. Then the discrepancy of $\mathcal{F}$ is*

$$\operatorname{disc}(\mathcal{P}(X)) = \left\lceil \frac{n}{2} \right\rceil.$$

Proof. Recall that the discrepancy of a set system $\mathcal{F}$ on a ground set X is defined by

$$\operatorname{disc}(\mathcal{F}) = \min_{\chi\colon X \to \{\pm 1\}} \max_{S \in \mathcal{F}} \left| \sum_{x \in S} \chi(x) \right|,$$

where the minimum is taken over all two-colorings of X, and the maximum over all sets in the system.

We prove the two requisite inequalities separately.

Lower bound. Fix an arbitrary coloring $\chi\colon X \to \{\pm 1\}$, and put

$$P = \{x \in X : \chi(x) = +1\}, \qquad N = \{x \in X : \chi(x) = -1\}.$$

Let $p = |P|$ and $q = |N|$, so $p + q = n$. Since $\mathcal{P}(X)$ contains every subset of X, it contains P and N in particular. Therefore

$$\max_{S \in \mathcal{P}(X)} \left| \sum_{x \in S} \chi(x) \right| \geq \max \left\{ \left| \sum_{x \in P} \chi(x) \right|, \left| \sum_{x \in N} \chi(x) \right| \right\} = \max\{p, q\}.$$

As $p + q = n$, we have $\max\{p, q\} \geq \lceil n/2 \rceil$. Since the coloring χ was arbitrary, it follows that

$$\operatorname{disc}(\mathcal{P}(X)) \geq \left\lceil \frac{n}{2} \right\rceil.$$

Upper bound. It remains to exhibit a coloring whose maximum discrepancy over all subsets is at most $\lceil n/2 \rceil$. Choose a coloring χ for which

$$|P| = \left\lceil \frac{n}{2} \right\rceil, \qquad |N| = \left\lfloor \frac{n}{2} \right\rfloor.$$

Such a coloring evidently exists. Now let $S \subseteq X$ be arbitrary, and write $S = S_+ \cup S_-$, where $S_+ = S \cap P$ and $S_- = S \cap N$. Then

$$\sum_{x \in S} \chi(x) = |S_+| - |S_-|.$$

Since $0 \leq |S_+| \leq \lceil n/2 \rceil$ and $0 \leq |S_-| \leq \lfloor n/2 \rfloor$, we obtain

$$-\left\lfloor \frac{n}{2} \right\rfloor \leq |S_+| - |S_-| \leq \left\lceil \frac{n}{2} \right\rceil.$$

Hence

$$\left|\sum_{x\in S}\chi(x)\right| \le \left\lceil \frac{n}{2} \right\rceil$$

for every $S \subseteq X$. Taking the maximum over $S \in \mathcal{P}(X)$ and then the minimum over colorings, we conclude that

$$\operatorname{disc}(\mathcal{P}(X)) \le \left\lceil \frac{n}{2} \right\rceil .$$

Combining the lower and upper bounds yields

$$\operatorname{disc}(\mathcal{P}(X)) = \left\lceil \frac{n}{2} \right\rceil . \qquad \blacksquare$$

□